

HOSPITAL MANAGEMENT SYSTEM

Project Report

Submitted By

Name: Rishi Prakash Soni

Roll No.: 2501220100192

Branch: B.Tech Computer Science & Engineering

College: Shri Ramswaroop Memorial College of Engineering and Management (SRMCEM)

Certificate

This is to certify that the project titled "Hospital Management System" has been successfully completed by Rishi Prakash Soni (Roll No. 2501220100192) in partial fulfillment of the requirements for the award of the Bachelor of Technology degree in Computer Science & Engineering at Shri Ramswaroop Memorial College of Engineering and Management (SRMCEM). The project work is original and has been carried out under proper guidance.

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Abstract

The Hospital Management System is a Python and MySQL based application developed to simplify and automate the management of hospital activities. The system provides facilities to manage patient records, doctor details, appointments, billing, and reports efficiently. It reduces paperwork, minimizes human errors, and improves the speed and accuracy of hospital operations. The project demonstrates the practical implementation of database management concepts using Python and MySQL.

Introduction

Hospitals deal with a large amount of information every day, including patient records, doctor details, appointments, medicines, and billing information. Managing these records manually is time-consuming and increases the possibility of errors. To overcome these problems, a computerized Hospital Management System has been developed.

This project allows hospital staff to maintain all records digitally. It provides an easy interface for entering, updating, deleting, and retrieving information whenever required. The system ensures that all data is stored securely in a MySQL database and can be accessed quickly.

Problem Statement

Traditional hospital management systems depend heavily on paper-based records, which consume time, require more storage space, and are difficult to maintain. Manual processes may also lead to data duplication and errors.

The objective of this project is to develop a computerized system that efficiently manages hospital information and improves the quality of healthcare services.

Objectives

Maintain patient records digitally.

Store doctor information.

Manage hospital appointments.

Generate patient bills automatically.

Reduce paperwork.

Improve efficiency and accuracy.

Provide quick access to information.

Existing System

The traditional system stores patient information manually in registers and files. Searching for old records takes time and there is a high possibility of losing important documents. Billing and appointment scheduling are also performed manually, making the process slow and inefficient.

Proposed System

The proposed Hospital Management System replaces manual work with a computerized application. The system stores all records in a MySQL database and provides options to add, update, delete, and search records easily. It also manages appointments and billing automatically.

Technologies Used

Programming Language

Python

Database

MySQL

SQL

Used for creating tables and managing records.

IDE

Visual Studio Code / PyCharm

Libraries

mysql-connector-python

Python Standard Library

Hardware Requirements

Intel Core i3 or above

4 GB RAM (Minimum)

500 GB Storage

Windows 10/11

Software Requirements

Python 3.x

MySQL Server

Visual Studio Code / PyCharm

MySQL Connector

Project Modules

1. Patient Management

This module allows users to add new patients, update existing information, delete patient records, and search patient details quickly.

Functions

Add Patient

Update Patient

Delete Patient

Search Patient

2. Doctor Management

This module stores doctor information such as name, specialization, contact details, and availability.

Functions

Add Doctor

Update Doctor

Delete Doctor

Search Doctor

3. Appointment Management

Patients can book appointments with doctors. The appointment details are stored in the database for future reference.

Functions

Book Appointment

View Appointments

Cancel Appointment

4. Billing Module

The billing module calculates patient charges and generates hospital bills automatically.

Functions

Generate Bill

Display Bill

Save Bill Record

5. Reports Module

This module generates reports related to patients, doctors, appointments, and billing records.

Database Design

The project uses MySQL as the backend database.

Main Tables

Patient

Doctor

Appointment

Billing

Each table stores related information and is connected through primary and foreign keys wherever required.

Working of the Project

User starts the application.

Main menu appears.

User selects a module.

Information is entered.

Data is validated.

Data is stored in the MySQL database.

User can update, search, or delete records.

Reports and bills are generated whenever required.

Advantages

User-friendly interface.

Fast processing.

Secure database storage.

Less paperwork.

Easy report generation.

Better hospital management.

Accurate patient records.

Limitations

Desktop application only.

Requires MySQL installation.

Single-user system.

Basic security features.

Future Scope

The project can be improved by adding:

Online appointment booking.

Patient login portal.

Doctor login system.

Email and SMS notifications.

Cloud database support.

Mobile application.

QR Code based patient identification.

Online payment gateway.

AI-based appointment prediction.

Testing

The application was tested using different patient and doctor records. Various operations such as insertion, updation, deletion, searching, billing, and report generation were verified successfully. The database stored all records correctly without data loss.

Conclusion

The Hospital Management System successfully automates various hospital operations including patient management, doctor management, appointment scheduling, billing, and report generation. The system reduces manual work, increases efficiency, improves data accuracy, and provides better management of hospital information. This project demonstrates the practical implementation of Python programming and MySQL database concepts in a real-world application.

References

[Python Official Documentation](#)

[MySQL Official Documentation](#)

[Python MySQL Connector Documentation](#)

[SQL Documentation](#)

[Visual Studio Code Documentation](#)

[Course Notes and Project Source Code](#)